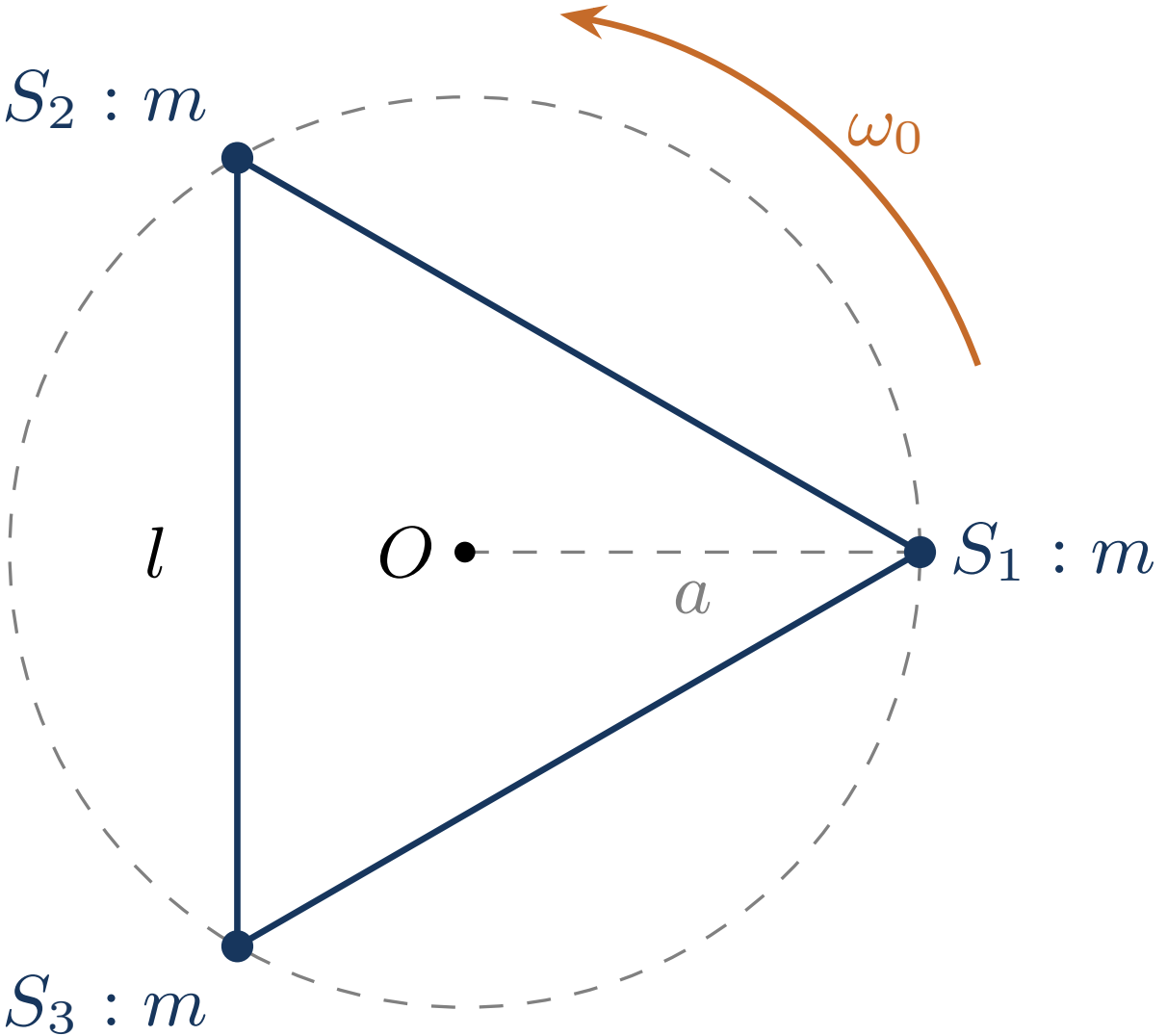
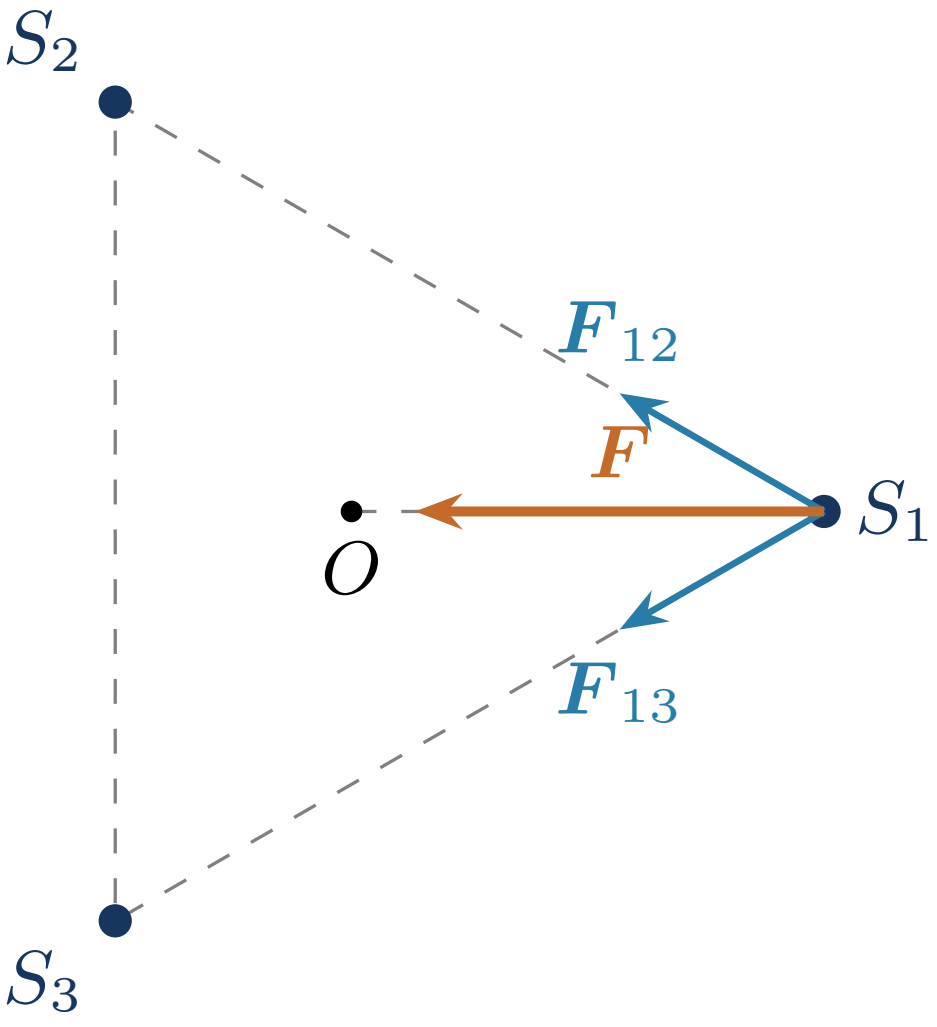
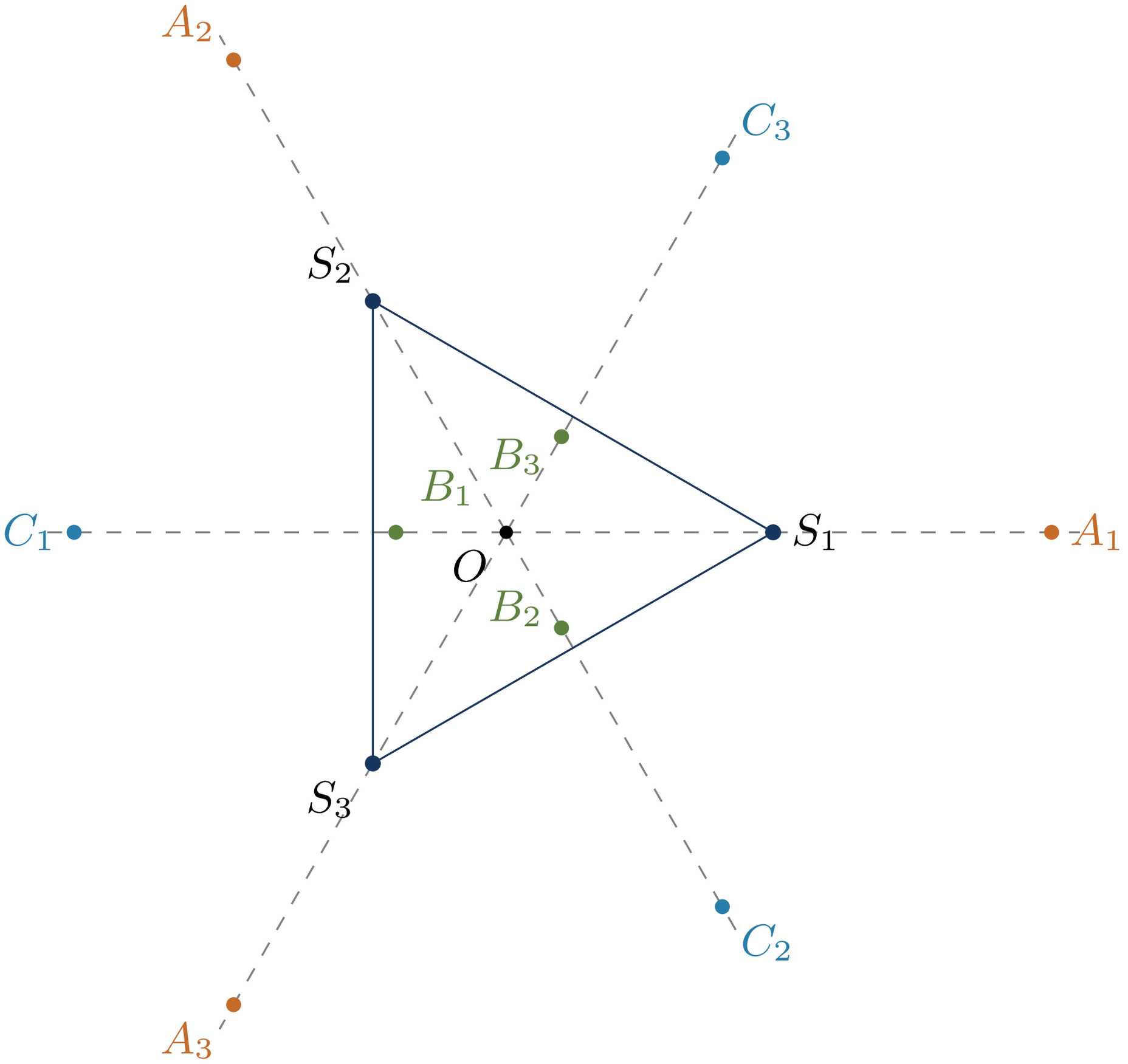

$S_2 : m$
ω_0
l
O
a
$S_1 : m$
$S_3 : m$

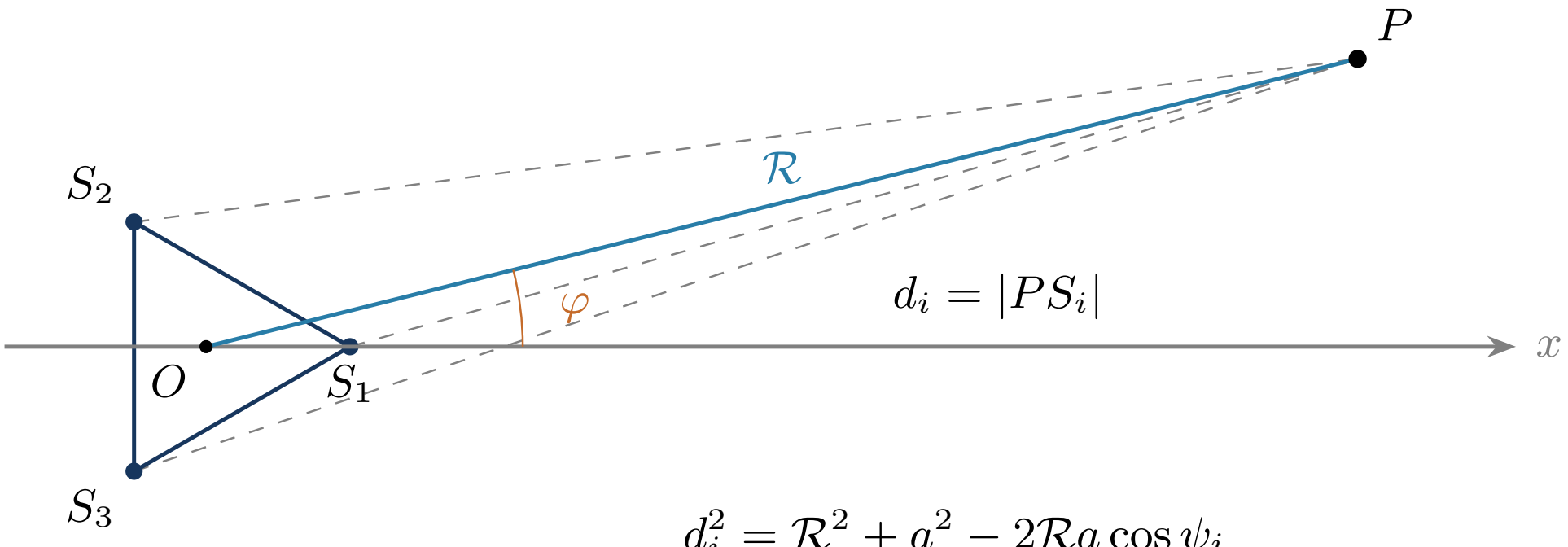


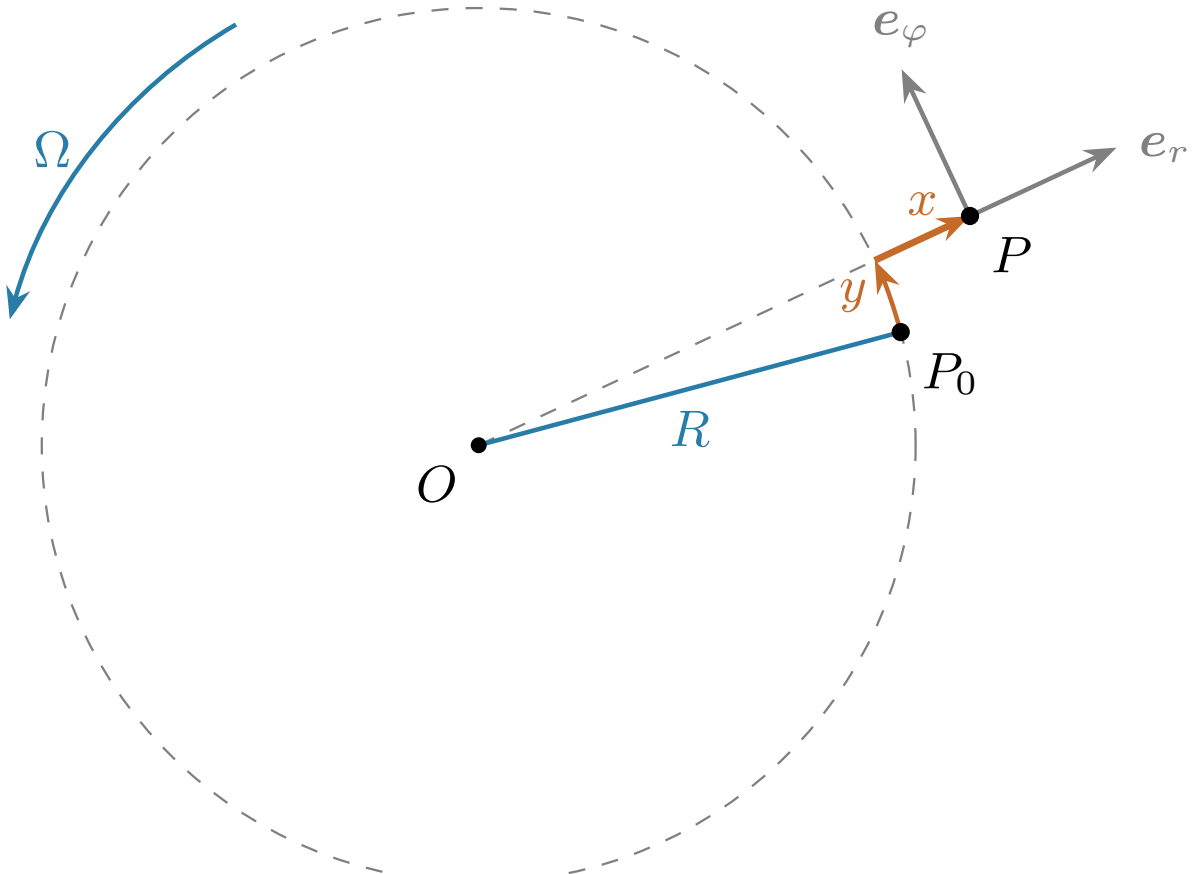
$F_{12} = F_{13} = Gm^2/l^2$

两力夹角 $60°$

合力指向共同质心 O


A_2
C_3
S_2
B_3
B_1
C_1
S_1
A_1
O
B_2
S_3
C_2
A_3


P
S_2
$\mathcal{R}$
φ
$d_i = |PS_i|$
x
O
S_1
S_3
$d_i^2 = \mathcal{R}^2 + a^2 - 2\mathcal{R}a\cos\psi_i$

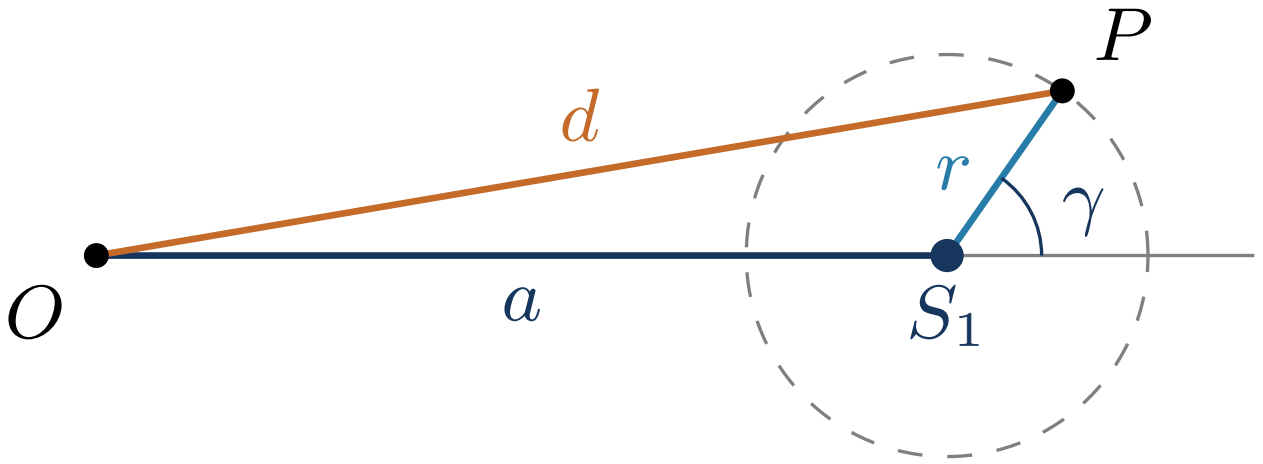


$\mathcal{R} = R + x,\ \varphi = \Omega t + y/R$

x：径向微小位移

y：切向微小位移

示意图将 x, y 放大显示



$$d^2 = a^2 + r^2 + 2ar\cos\gamma\,,\;\; r \ll l$$